National University
of computer and emerging sciences

Hospital Master

Final Report

Fundamentals of Software Engineering

Due Date: 7-Dec-2023

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Section	CY-A
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Hospital Master

Introduction:

The project revolves around the creation of a Hospital Management System, aimed at solving the pressing issue of inefficient healthcare administration. This problem holds immense importance as it directly affects the quality of patient care and overall operational efficiency within healthcare facilities. Our chosen solution entails the development of a robust software platform that automates critical tasks including patient registration, appointment scheduling, billing, and medical record management. By implementing this system, we aim to streamline hospital operations, enhance resource allocation, and ultimately improve patient experiences, addressing the pressing needs of modern healthcare institutions.

User Stories:

Sprint 1:

1. As a patient, I want to be able to register for an account so that I can access the hospital's services online.

   Tasks:

   Design user registration form

   Implement user registration functionality

   Create a database schema for patient accounts

2. As a doctor or nurse, I want to be able to log in to my account securely.

   Tasks:

   Design login interface

   Implement login functionality with authentication

   Develop a secure session management system

3. As a healthcare provider, I want to create electronic health records (EHRs) for patients, including their medical history and current conditions.

   Tasks:

   Design the EHR creation interface

   Implement EHR creation functionality

   Ensure validation and proper storage of EHR data

Sprint 2:

1. As a healthcare provider, I want to be able to update and edit EHRs to ensure accuracy.

   Tasks:

   - Design the EHR editing interface
   - Implement EHR editing functionality
   - Enforce proper access control for EHR updates

2. As a system administrator, I want to ensure that patient data in EHRs is securely encrypted to protect sensitive information.

   Tasks:

   - Research and select appropriate encryption algorithms
   - Implement data encryption and decryption mechanisms
   - Test encryption across the system to ensure data security

3. As a patient, I want to provide feedback about my experience with the hospital's services.

   Tasks:

   - Design a feedback submission form
   - Implement feedback submission functionality
   - Ensure data validation and storage of feedback

Sprint 3:

1. As a patient, I want to schedule appointments with doctors and nurses online.

   Tasks:

   - Design the appointment scheduling interface
   - Implement appointment scheduling functionality
   - Integrate appointment data with EHRs for healthcare providers' reference

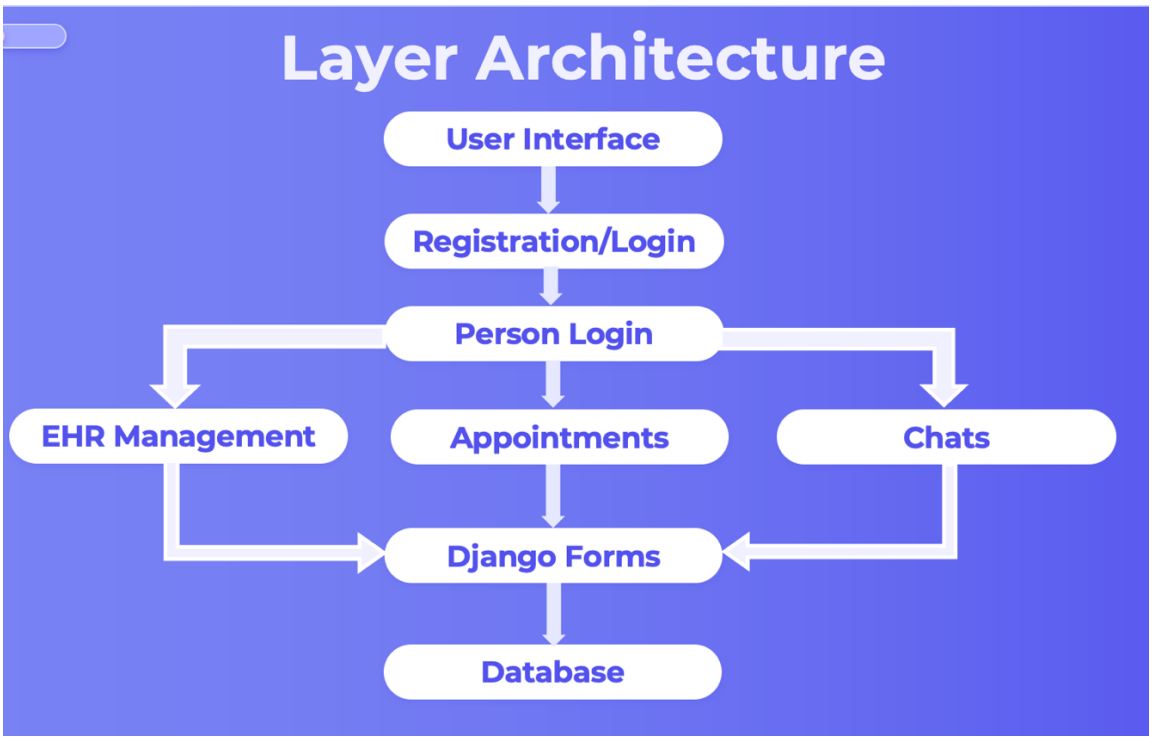
2. As a patient, I want the option to chat with my assigned doctor and hospital staff through the hospital's online portal for inquiries or support.

   Tasks:

   - Modify the chat system to allow patients to initiate chats with their assigned doctors and hospital staff.
   - Ensure that the chat system maintains patient privacy and security, restricting access to authorized personnel only.
   - Implement a notification system for incoming patient chats, ensuring doctors and staff are promptly alerted to new patient inquiries and messages.

Architecture:

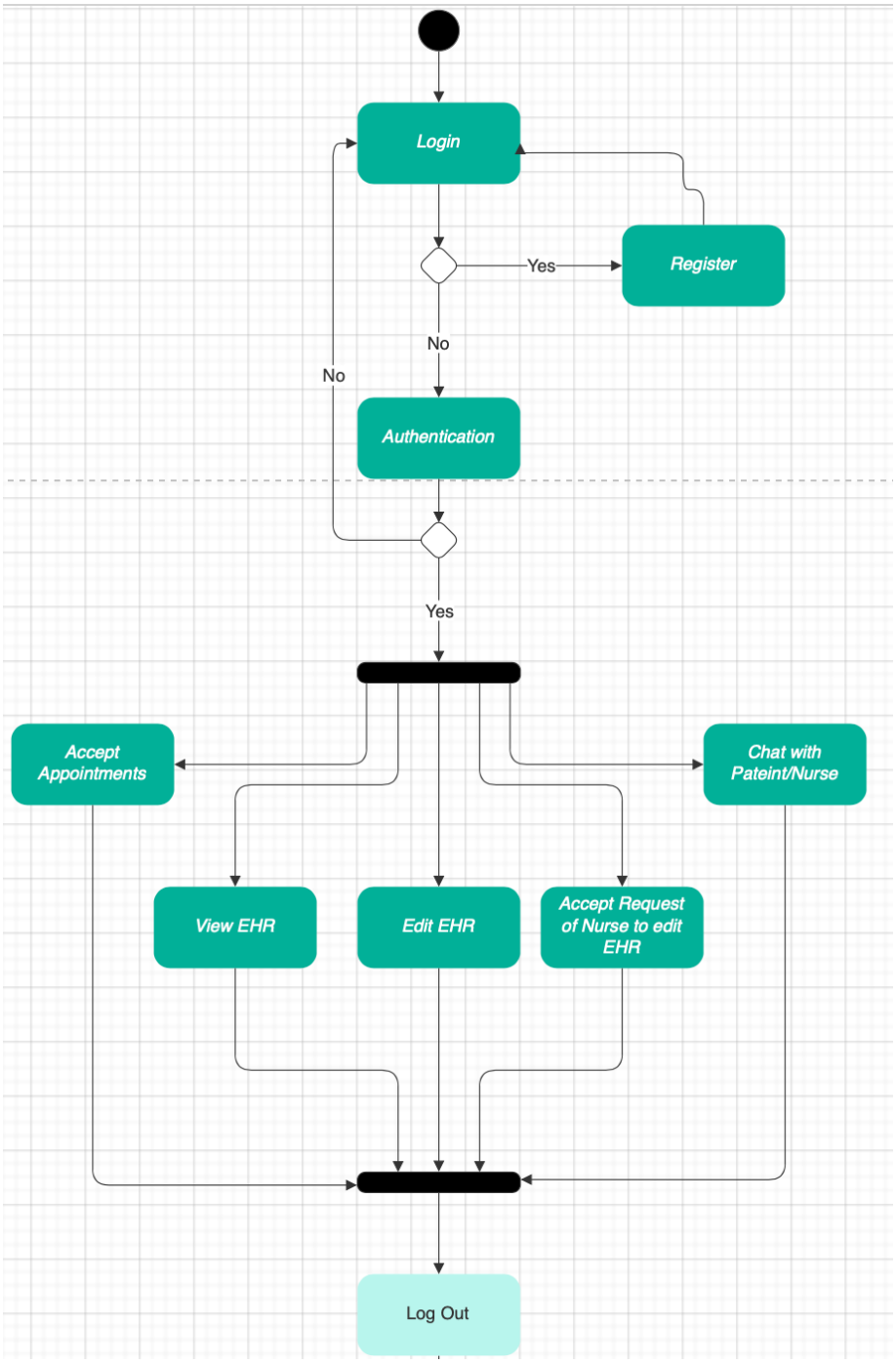
We have identified layered architecture in our project.



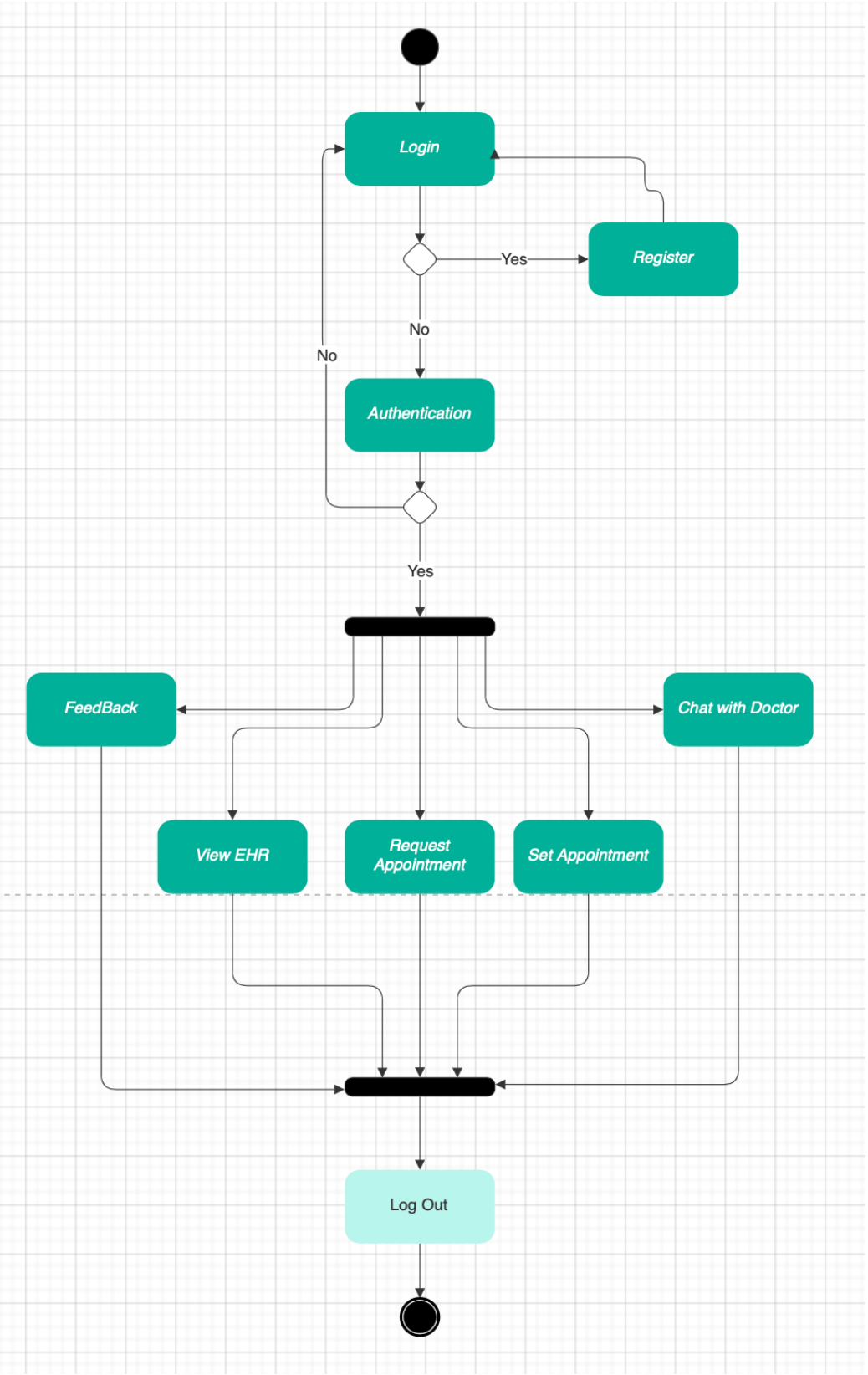
Design Artifacts:

Activity Diagrams:

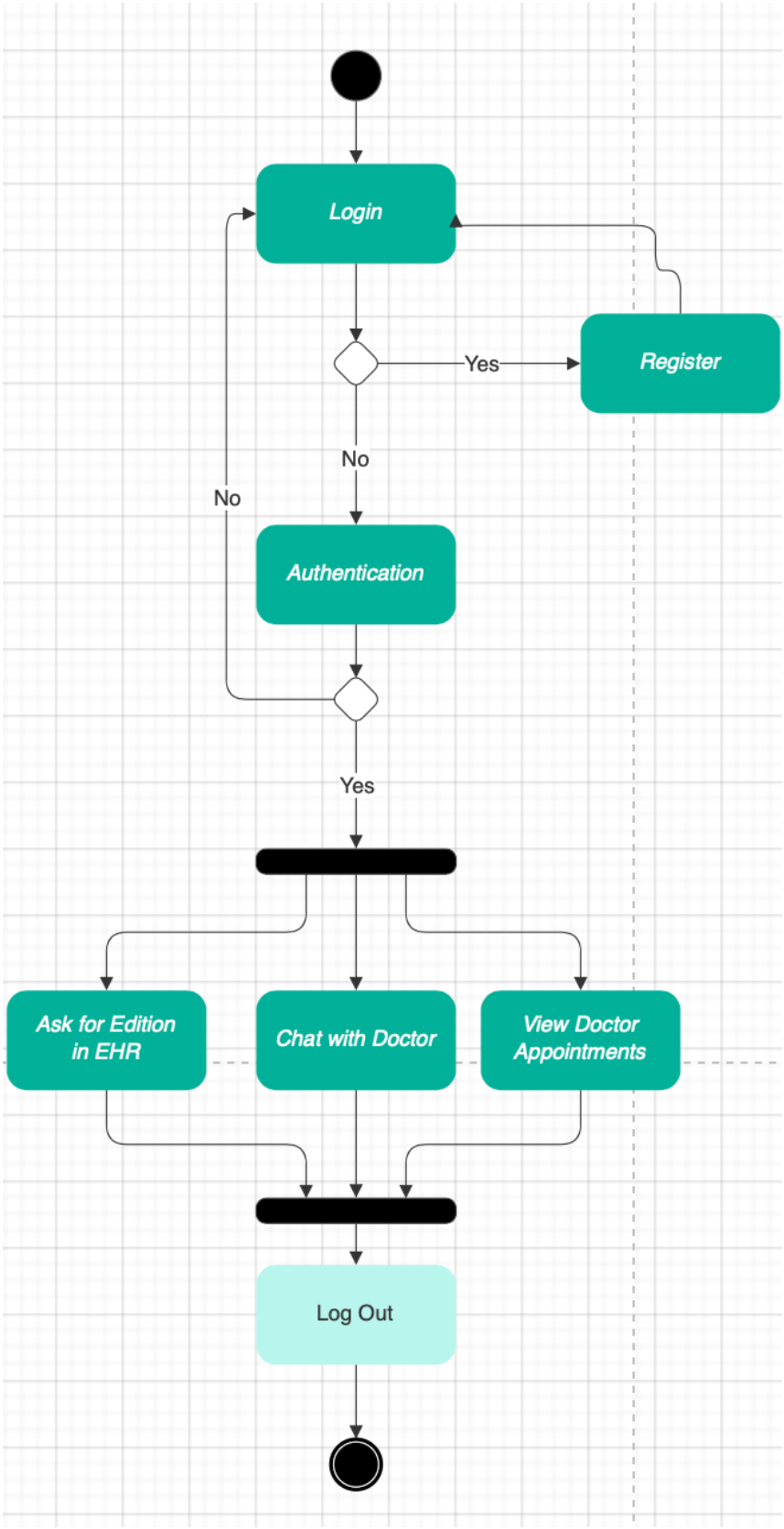
Doctor Activity Diagram:



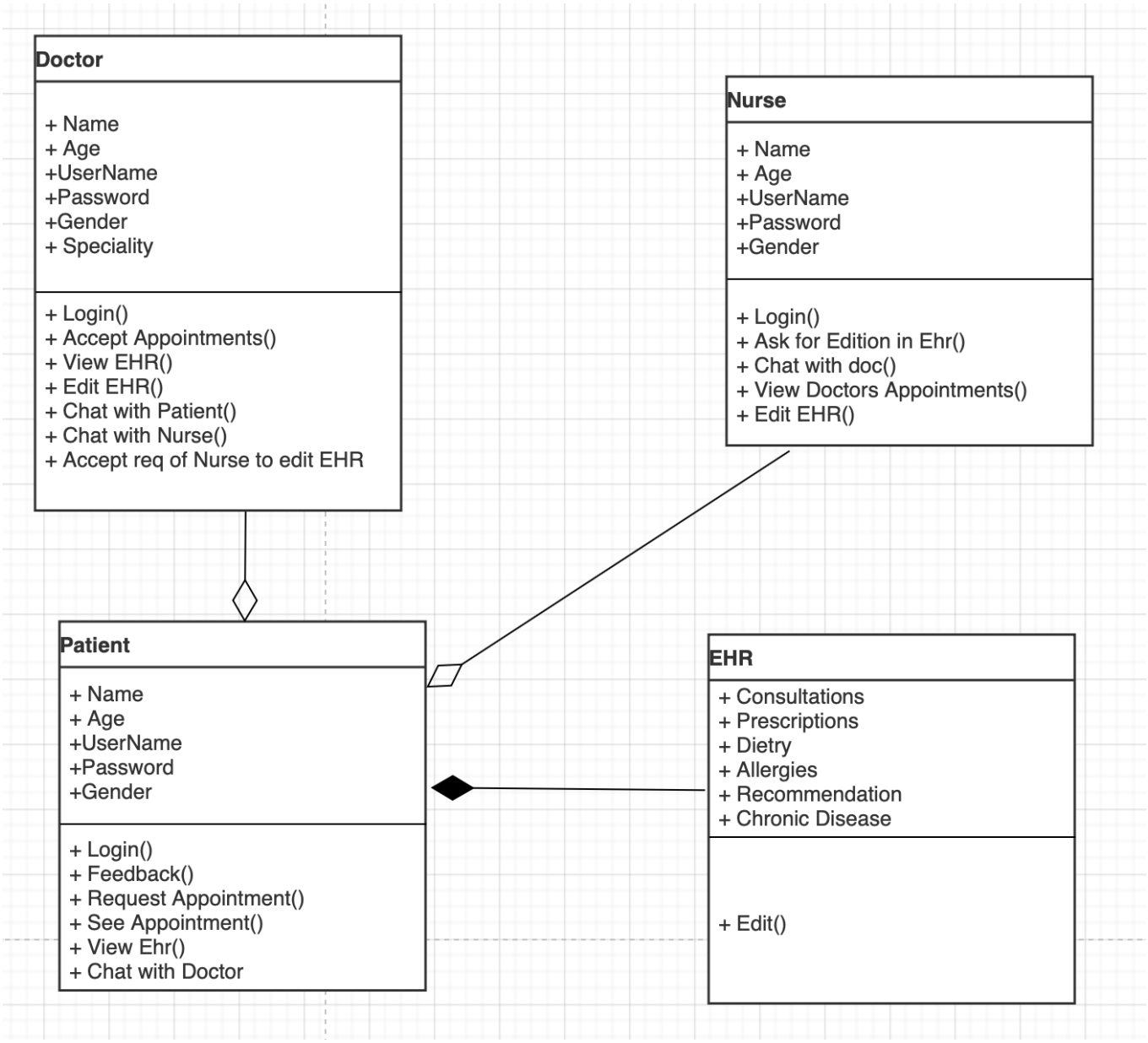
Patient Activity Diagram:



Nurse Activity Diagram:

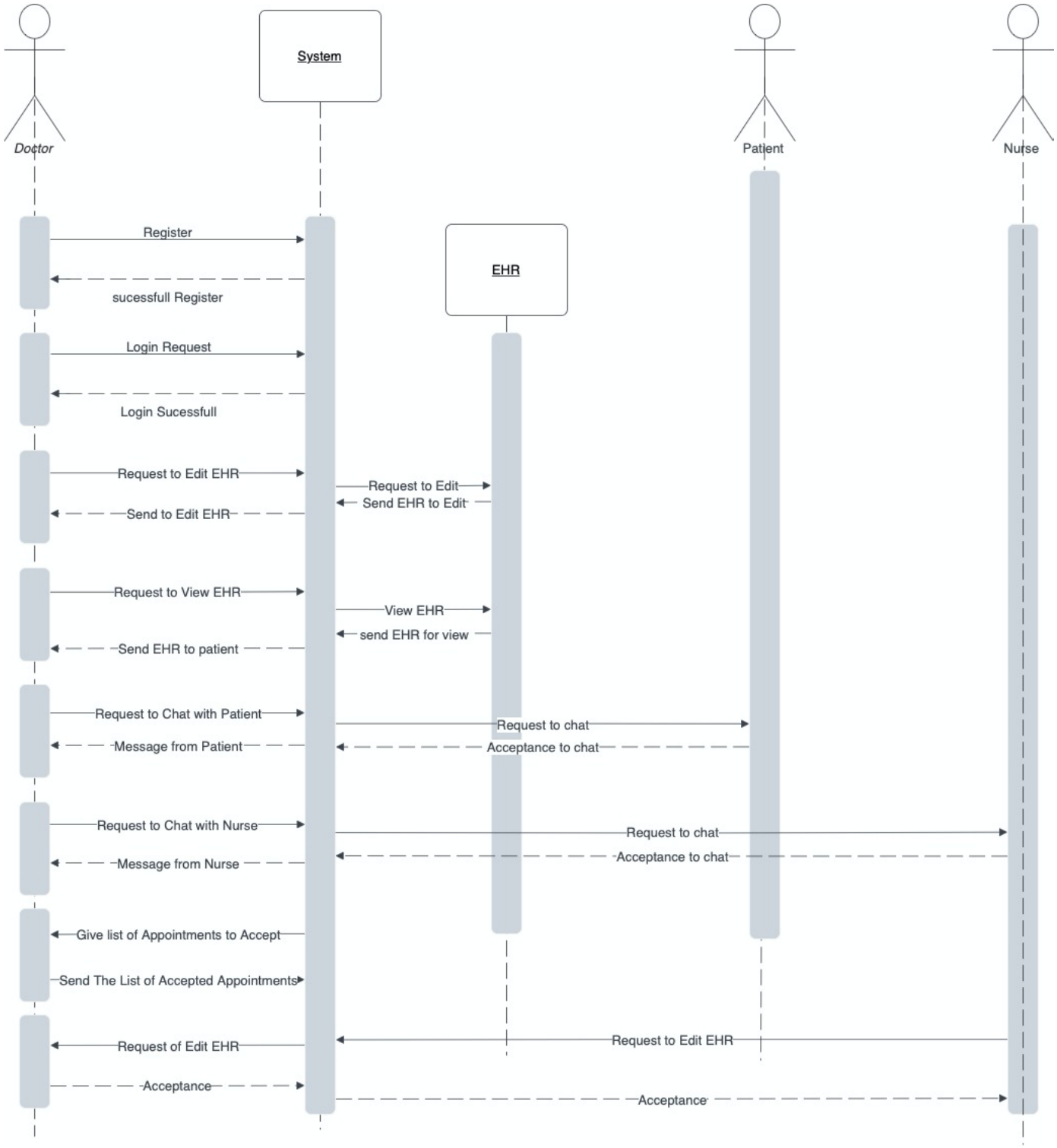


Class Diagram:

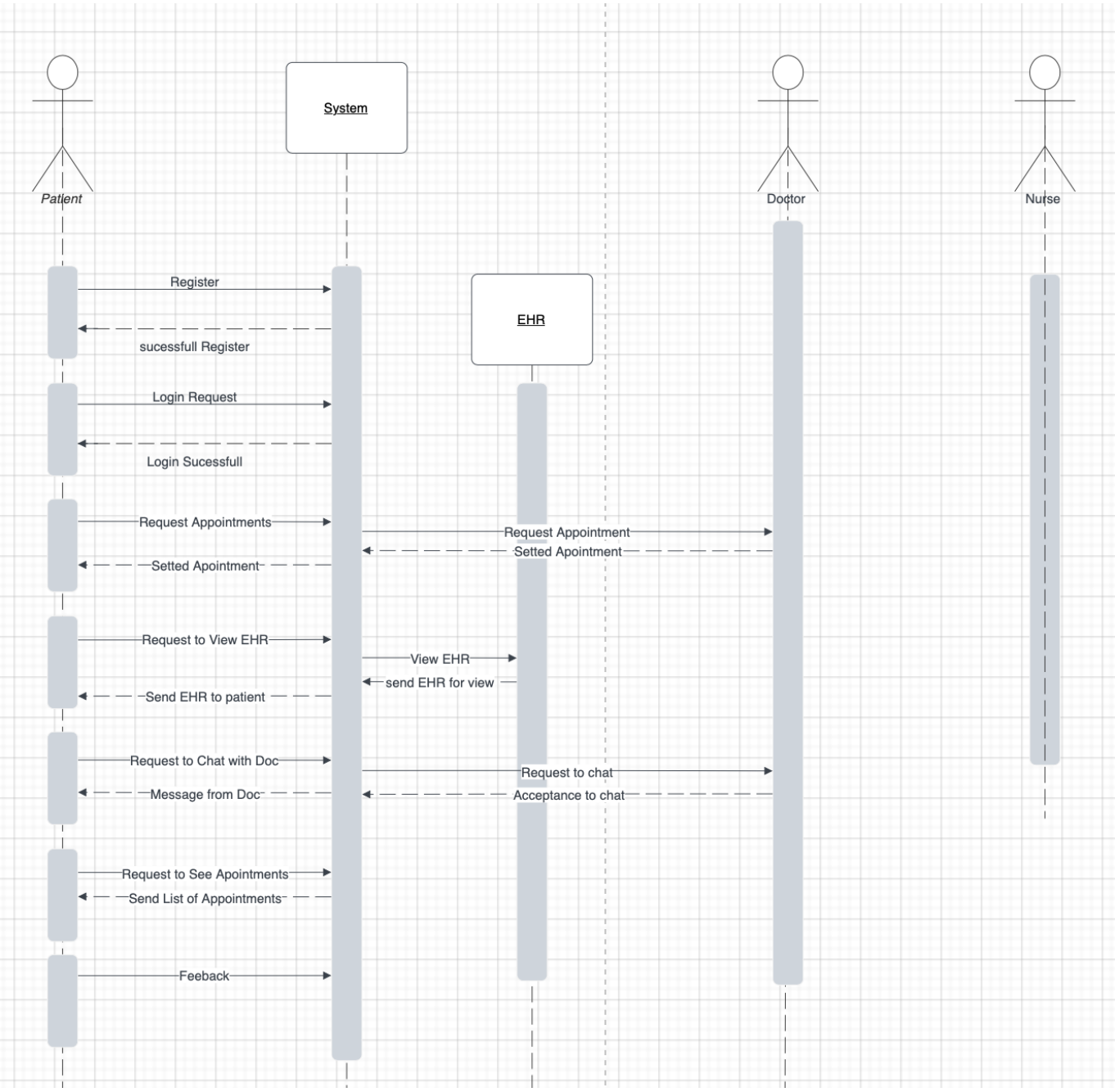


Sequence Diagrams:

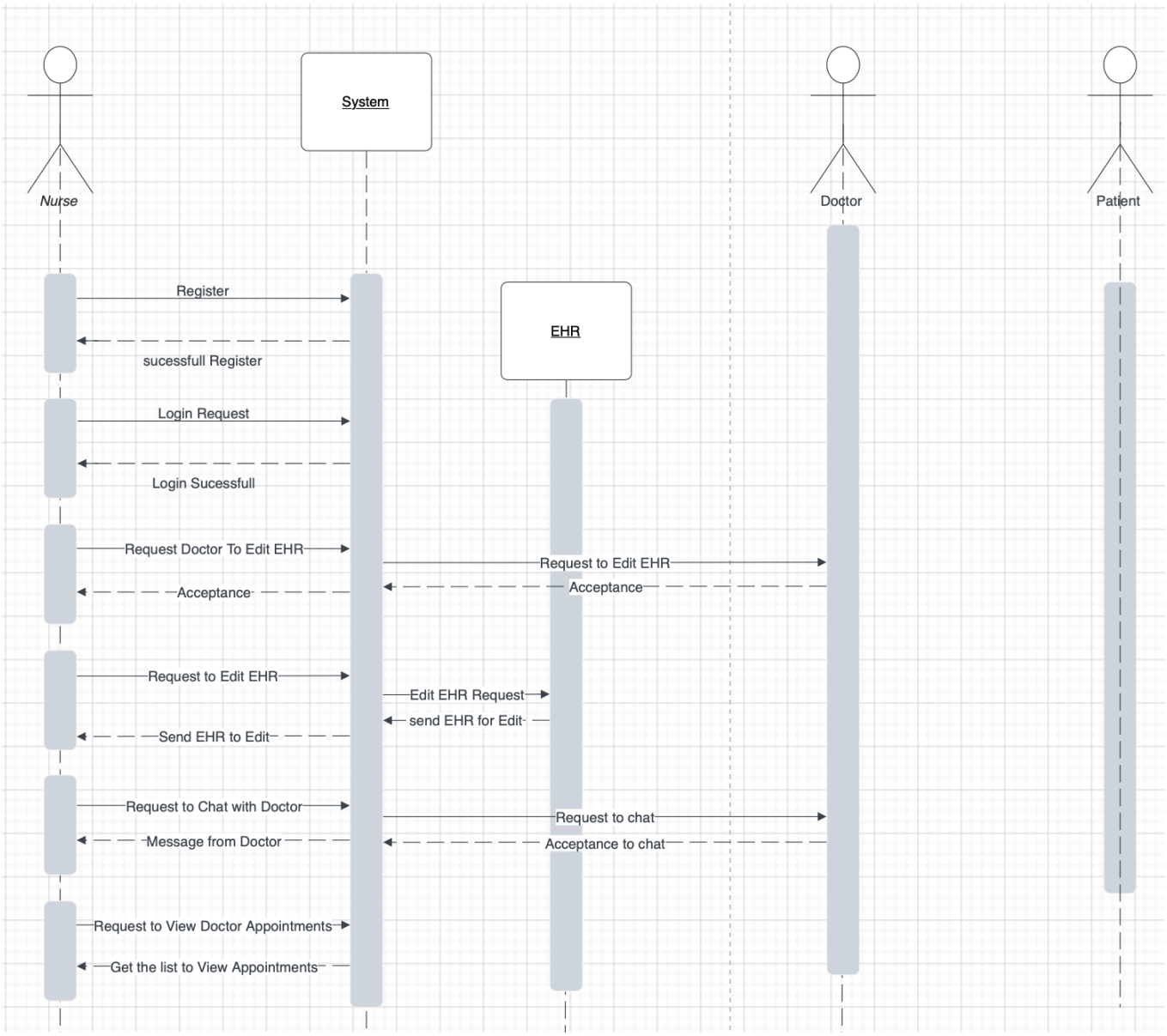
Doctor Sequence Diagram:



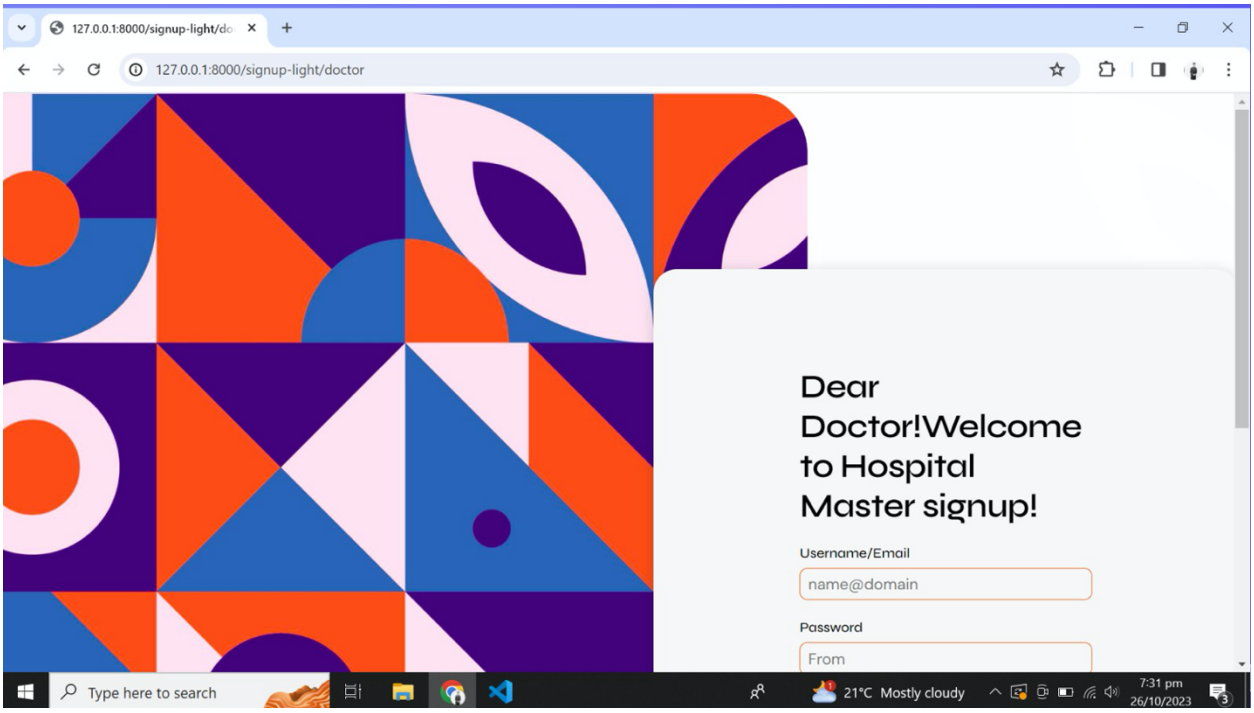
Patient Sequence Diagram:

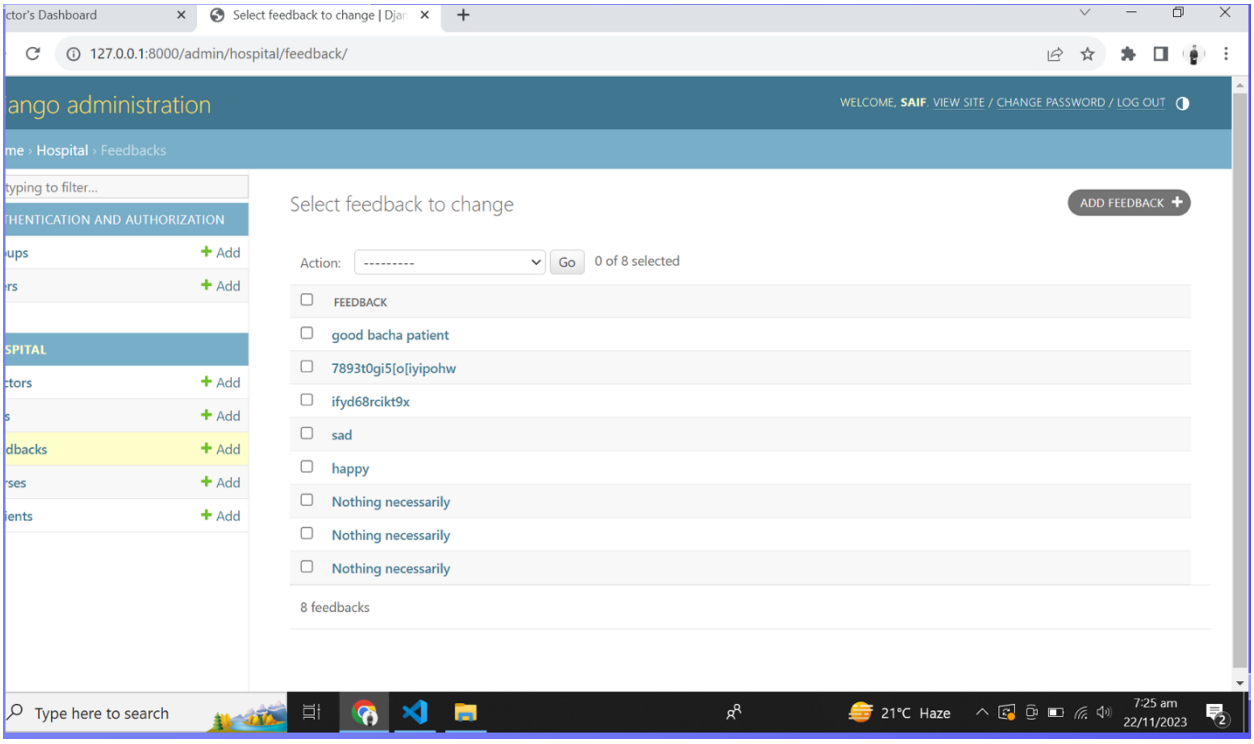


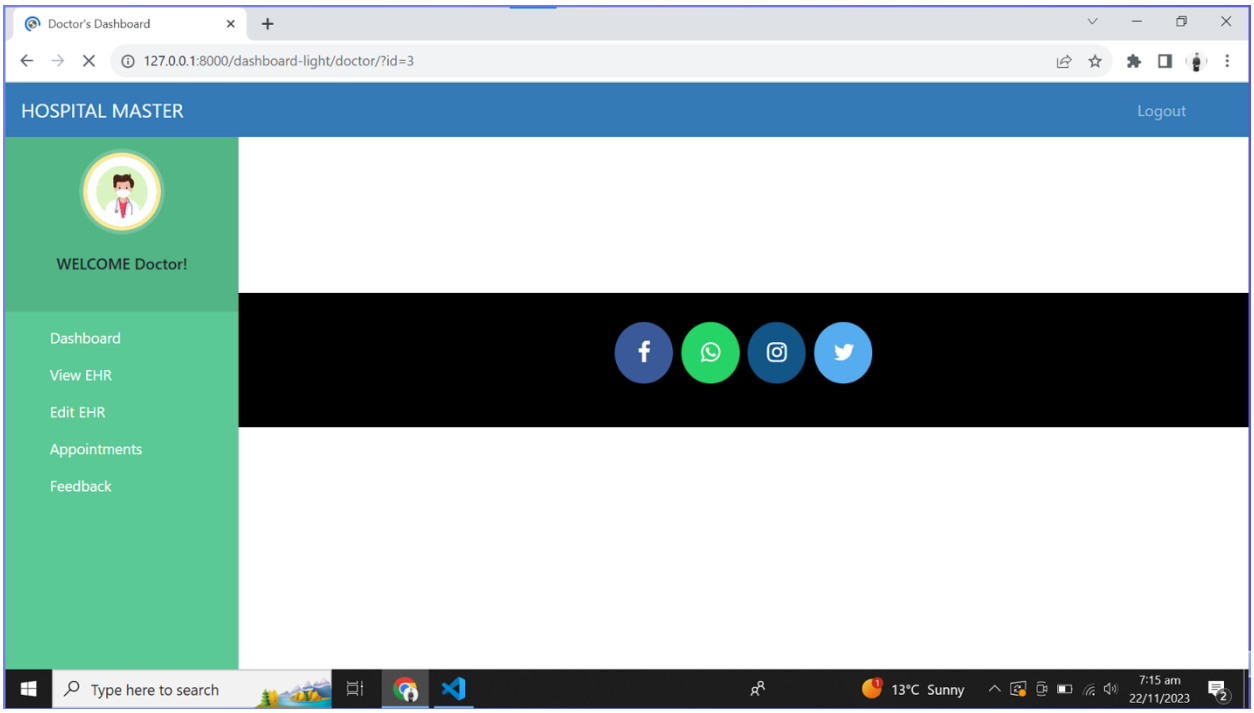
Nurse Sequence Diagram:



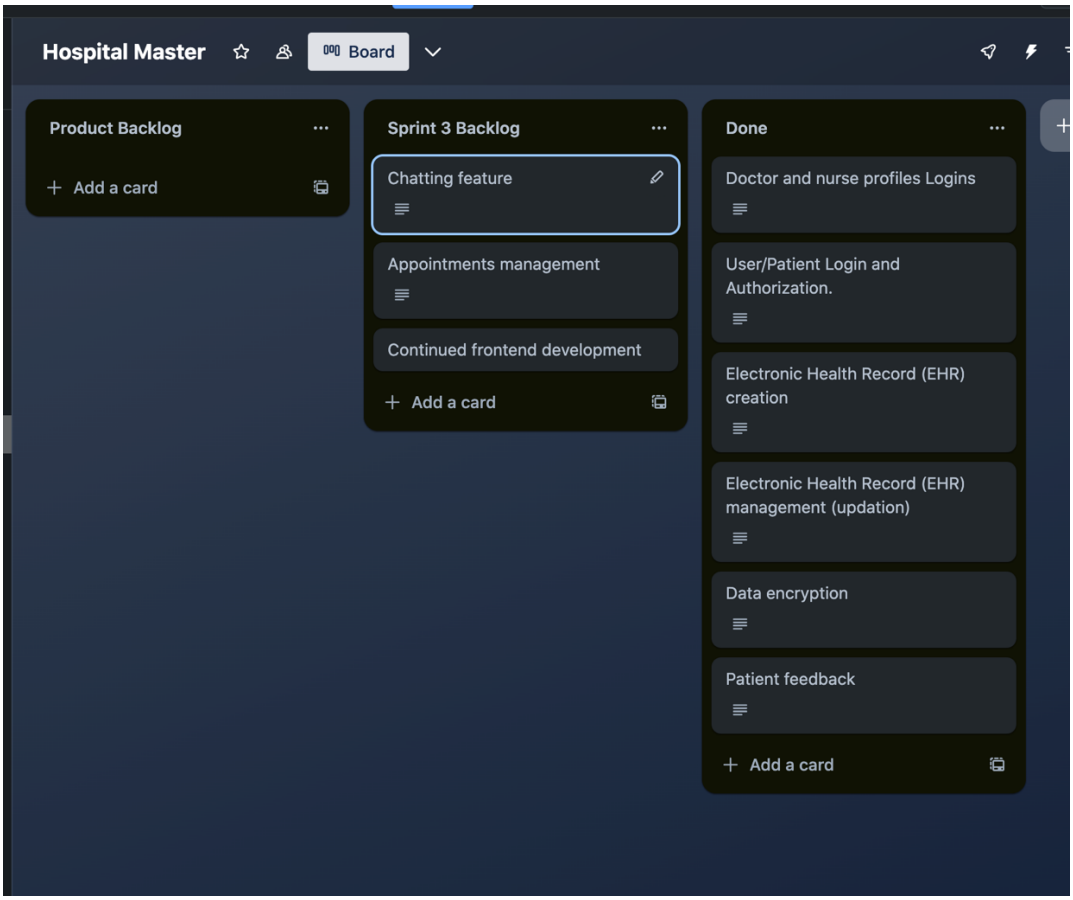
Implementation Screenshots:

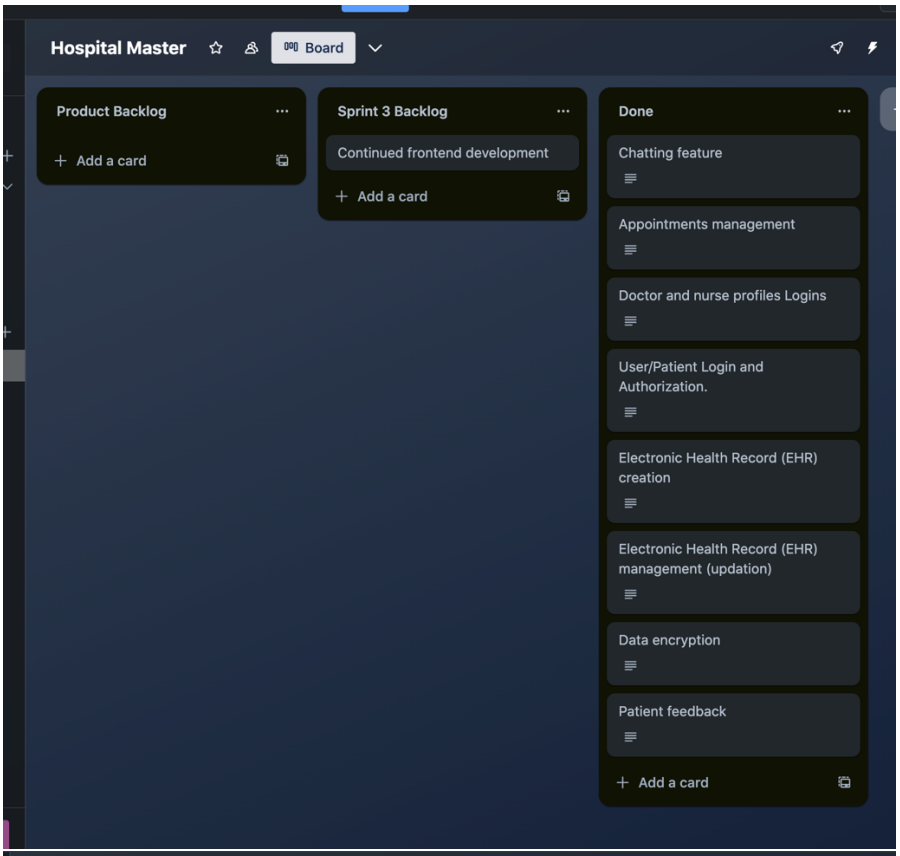


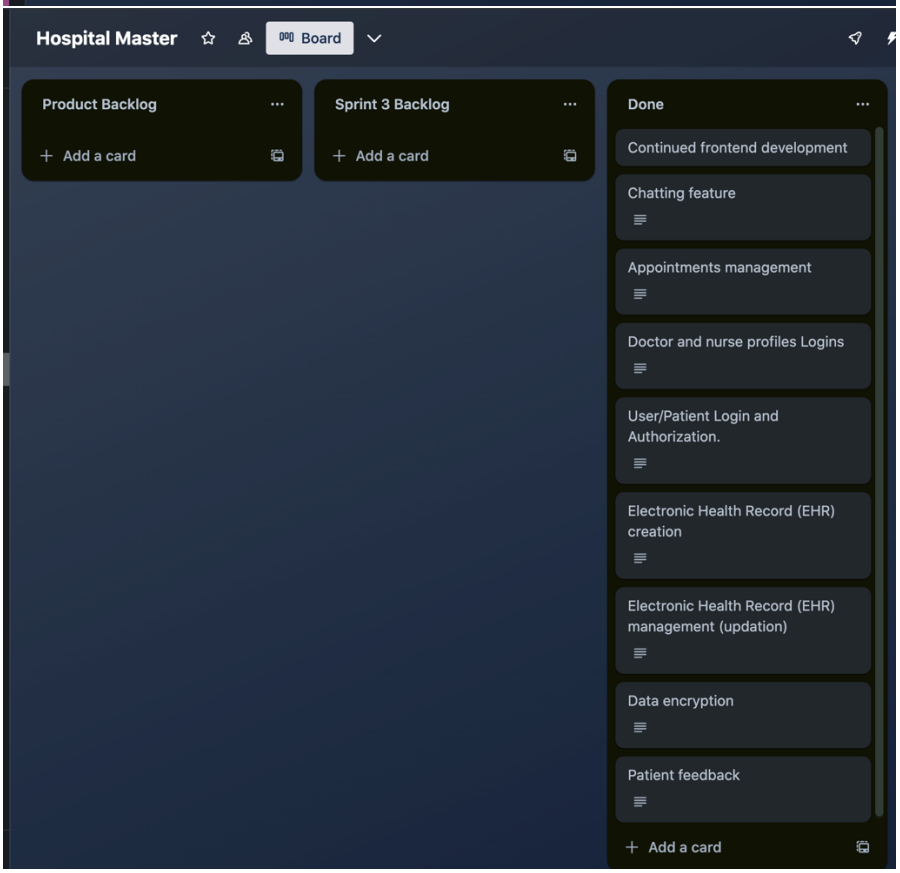




Trello Board Screenshots:




Hospital Master
Board
Product Backlog
Add a card
Sprint 3 Backlog
Continued frontend development
Add a card
Done
Chatting feature
Appointments management
Doctor and nurse profiles Logins
User/Patient Login and Authorization.
Electronic Health Record (EHR) creation
Electronic Health Record (EHR) management (updation)
Data encryption
Patient feedback
Add a card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Work Division:

Saif-ur-Rehman	Humayun Masood	Umer Farooq
❑ Database management ❑ EHR Creation ❑ Appointment Approvals ❑ Dynamic Styling ❑ Chat system	❖ Login Management ❖ EHR Updation ❖ Appointments requests ❖ Design Artifacts ❖ Static Styling	✓ Frond-End management ✓ Patient Feedback Form ✓ Documentation ✓ Trello board management